

MRXtra: An AI-Assisted System for MRI Safety Screening, Workflow Decision Support, and Protocol Optimization in Resource-Limited Settings

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Purpose: MRI is a mainstay of diagnostic imaging whose use is spreading in Africa. But MRI poses substantial safety hazards when screening is inadequate. This risk is particularly acute in Africa, where paper-based workflows, overburdened care teams, and sparse resources elevate human error rates, strain operational efficiency, and cause gaps in data capture. For example, implant safety database queries, standard in the global north, are not available in much of Africa. Notably, AI work in radiology has focused on image reconstruction and interpretation, bypassing the pre-scan phase. Therefore our project addresses this safety and data quality gap by embedding AI and digital processes at the point of screening and protocol selection, before the patient enters the scanner. The project's key deliverable is an AI-supported screening protocol that is tailored for practical use in Africa, to improve patient safety and data capture quality.

Materials and Methods: Based on a questionnaire to Nigerian MRI radiographers (n=150), we developed MRXtra, a web app with a structured SQL backend combining a lightweight LLM layer for natural language interpretation, a deterministic rule-based inference engine, and plain-language explanation generation. The rule-based core encodes 42 evidence-based decision rules mapped to ACR 2024 across eight clinical domains, ensuring interpretability. The first LLM operates as a suggestion layer feeding into the deterministic tree, without overriding safety classifications. The second LLM generates patient-specific protocol and acquisition parameters within established safety constraints. Functionalities include adaptive questioning, real-time contraindication flagging, structured safety summaries, protocol optimization, and an implant safety database query module. Initial usability evaluation with radiographers assessed workflow integration. Two blinded independent clinical experts (radiologist and radiographer) rated all 42 rules for correctness (1 = Correct, 0 = Incorrect/Partial) severity-weighted on a 1–5 scale (low-risk to life-threatening). Fifteen patient vignettes tested integrated rule interactions. Discrepancies were resolved through adjudication.

Results: Outcomes showed improved clarity and completeness of MRI safety data capture. Adaptive questioning reduced unnecessary inputs while ensuring critical safety information was captured. Real-time risk flagging, protocol optimization, action plan tracking, and workflow decision support were perceived as valuable differentiators. Participants reported improved efficiency, ease of use, compliance, and the value of structured electronic records. In the expert validation, 40/42 (95.2%) and 41/42 (97.6%) rules were rated correct by Raters 1 and 2, respectively. Severity-weighted correctness was 95.4% (R1) and 98.2% (R2). All seven life-threatening rules were rated correct by both experts. Cohen's kappa was 0.66 (substantial agreement). One safety-critical discrepancy was resolved through adjudication. Scenario testing across 15 vignettes showed 100% (R 1) and 93.3% (R 2) clinical appropriateness, with 14/15 agreements. Domain-specific correctness was 88.9% (Implant, 16/18), 100% (Ferromagnetic, 6/6), and 100% (Contrast, 6/6).

Conclusion: MRXtra addresses a serious safety gap for MRI use in Africa. It represents a paradigm shift from passive form-based screening to an interactive decision-support workflow. By embedding transparent, interpretable AI upstream before the patient reaches the scanner, it closes safety gaps that traditional paper-based screening often misses, especially in LMIC settings, while preserving the clinical reasoning process.

Medical/clinical significance: In routine practice, incomplete safety histories and variable staff expertise remain leading contributors to MRI adverse events. MRXtra addresses this by dynamically expanding data capture through adaptive questioning and real-time risk flagging, which directly reduces the likelihood of contraindicated scans and protocol errors. It focuses on the practical needs and constraints of resource-limited settings to standardize screening rigor, adds implant safety queries, and supports compliance with safety guidelines without requiring specialized MRI safety personnel at every touchpoint.

Prospect of application: Ethical and institutional approvals have been secured for evaluation and prospective clinical deployment at Medserve Kano Diagnostic Center in Aminu Kano Teaching Hospital. Ongoing work includes: prospective comparison with paper-based screening for workflow efficiency and safety compliance, TAM/SUS assessment, implant query pricing, and an image quality protocol module.

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